

Programming for AI

BS(AI) Part-2- 2nd Semester

Prof. Dr. Imtiaz Ali Korejo

Python Variables

Week-1 , Lecture-2





Outline of Today's Lesson

1. Python Comments

- a. Python Comments
- b. Creating A Comment
- c. Multiple Comments
- d. Multiline Comments

2. Python Variables

- a. Python Variables,
 - i. creating variables
 - ii. Get the Type
 - iii. Single or Double Quotes
 - iv. Case Sensitive
- b. Variable Names,
 - i. Variable Names and their Rules
 - ii. Multi Words Variable Name
 - iii. Camel Case
 - iv. Pascal Case

v. Snake Case

- c. Assign Multiple Values
 - i. Name Values to Multiple Variables
 - ii. Obe Value to Multiple Variables
 - iii. UnPacck a Collection





Python Comments

- Comments can be used to explain Python code.
- Comments can be used to make the code more readable.
- Comments can be used to prevent execution when testing code.
-





Creating a Comment

- Comments starts with a **#**, and Python will ignore them:

```
#This is a comment  
print("Hello, World!")
```

- Comments can be placed at the end of a line, and Python will ignore the rest of the line:

```
print("Hello, World!") #This is a comment
```





Multiline Comments

- Python does not really have a syntax for multiline comments.
- To add a multiline comment you could insert a `#` for each line:

```
#This is a comment  
#written in  
#more than just one line  
print("Hello, World!")
```

The screenshot shows a code editor window with a dark theme. The code is written in Python. The first three lines are comments, each starting with a green hash symbol (#). The fourth line is a print statement. The editor has a search bar at the top, a file explorer on the left, and a status bar at the bottom showing 'Ln 9, Col 1', 'Spaces: 4', 'UTF-8', 'LF', and 'JavaScript'.





Multiline string as Comment

- Since Python will ignore string literals that are not assigned to a variable, you can add a multiline string (triple quotes) in your code, and place your comment inside it:
- As long as the string is not assigned to a variable, Python will read the code, but then ignore it, and you have made a multiline comment.

```
"""  
This is a comment  
written in  
more than just one line  
"""  
  
print("Hello, World!")
```





Section-2

Python Programming

VARIABLES

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Constants

- Fixed values such as numbers, letters, and strings, are called “constants” because their value does not change
- Numeric constants are as you expect
- String constants use single quotes (') or double quotes (")

```
>>> print(123)
123
>>> print(98.6)
98.6
>>> print('Hello world')
Hello world
```





Variables

A variable is a named place in the memory where a programmer can store data and later retrieve the data using the variable “name”

Programmers get to choose the names of the variables

You can change the contents of a variable in a later statement

`x = 12.2`

`y = 14`

`x`

12.2

`y`

14





Variables

A variable is a named place in the memory where a programmer can store data and later retrieve the data using the variable “name”

Programmers get to choose the names of the variables

You can change the contents of a variable in a later statement

`x = 12.2`

`y = 14`

`x = 100`

`x`

~~12.2~~ 100

`y`

14





Creating Variables

- Variables are containers for storing data values.
- Python has no command for declaring a variable.
- A variable is created the moment you first assign a value to it.

```
x = 5  
y = "John"  
print(x)  
print(y)
```





- Variables do not need to be declared with any particular *type*, and can even change type after they have been set.

```
x = 4          # x is of type int
x = "Sally"    # x is now of type str
print(x)
```





Casting

- If you want to specify the data type of a variable, this can be done with casting.

```
x = str(3)      # x will be '3'  
y = int(3)      # y will be 3  
z = float(3)    # z will be 3.0
```





Get the Type

- You can get the data type of a variable with the **type()** function.

```
x = 5  
y = "John"  
print(type(x))  
print(type(y))
```

```
Output:\n<class 'int'>  
<class 'str'>
```





Single or Double Quotes?

- String variables can be declared either by using single or double quotes:

```
x = "John"  
# is the same as  
x = 'John'
```





Case-Sensitive

- Variable names are case-sensitive.

```
a = 4
```

```
A = "Sally"
```

```
#A will not overwrite a
```



Section-3

Python Programming

PYTHON - VARIABLE NAMES



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Variable Names

- A variable can have a short name (like x and y) or a more descriptive name (age, carname, total_volume).

Rules for Python variables:

- A variable name must start with a **letter** or the **underscore** character
- A variable name **cannot** start with a **number**
- A variable name can only contain **alpha-numeric characters** and **underscores** (A-z, 0-9, and _)
- Variable names are **case-sensitive** (age, Age and AGE are three different variables)
- A variable name cannot be any of the **Python keywords**.





Legal variable names:

```
myvar = "John"  
my_var = "John"  
_my_var = "John"  
myVar = "John"  
MYVAR = "John"  
myvar2 = "John"
```

💡 Remember that variable names are case-sensitive





Illegal variable names:

2myvar = "John"

my-var = "John"

my var = "John"





Reserved Words

You cannot use **reserved words** as variable names / identifiers

False	await	else	import	pass
None	break	except	in	raise
True	class	finally	is	return
and	continue	for	lambda	try
as	def	from	nonlocal	while
assert	del	global	not	with
async	elif	if	or	yield





Multi Words Variable Names

- Variable names with more than one word can be difficult to read.
- There are several techniques you can use to make them more readable:

Camel Case

- Each word, except the first, starts with a capital letter:

```
myVariableName = "John"
```





Multi Words Variable Names

Pascal Case

- Each word starts with a capital letter:

```
MyVariableName = "John"
```

Snake Case

- Each word is separated by an underscore character:

```
my_variable_name = "John"
```





Python Variable Name Rules

- Must start with a letter or underscore _
- Must consist of letters, numbers, and underscores
- Case Sensitive

Good: spam eggs spam23 _speed

Bad: 23spam #sign var.12

Different: spam Spam SPAM





Assign Multiple Values

- Python allows you to assign values to multiple variables in one line:

```
x, y, z = "Orange", "Banana", "Cherry"  
print(x)  
print(y)  
print(z)
```

OutPut:

```
Orange  
Banana  
Cherry
```

Note: Make sure the number of variables matches the number of values, or else you will get an error.





One Value to Multiple Variables

- And you can assign the *same* value to multiple variables in one line:

```
x = y = z = "Orange"  
print(x)  
print(y)  
print(z)
```





Unpack a Collection

- If you have a collection of values in a list, tuple etc. Python allows you to extract the values into variables. This is called *unpacking*.

```
fruits = ["apple", "banana", "cherry"]  
x, y, z = fruits  
print(x)  
print(y)  
print(z)
```

```
Output:  
apple  
banana  
cherry
```





A Good Practice

- Since we programmers are given a choice in how we choose our variable names, there is a bit of “best practice”





```
x1q3z9ocd = 35.0  
x1q3z9afd = 12.50  
x1q3p9afd = x1q3z9ocd * x1q3z9afd  
print(x1q3p9afd)
```

What is this bit
of code doing?





```
x1q3z9ocd = 35.0  
x1q3z9afd = 12.50  
x1q3p9afd = x1q3z9ocd * x1q3z9afd  
print(x1q3p9afd)
```

```
a = 35.0  
b = 12.50  
c = a * b  
print(c)
```

What are these bits
of code doing?





```
x1q3z9ocd = 35.0  
x1q3z9afd = 12.50  
x1q3p9afd = x1q3z9ocd * x1q3z9afd  
print(x1q3p9afd)
```



```
a = 35.0  
b = 12.50  
c = a * b  
print(c)
```



What are these
bits of code doing?

```
hours = 35.0  
rate = 12.50  
pay = hours * rate  
print(pay)
```





LEARNING RESOURCES

Source Reference used in Slide

- <https://www.w3schools.com/python>

Other Resources

- <https://realpython.com/>
- <https://docs.python.org/3/>
- <https://docs.python.org/3/tutorial/index.html>
- <https://www.tutorialspoint.com/python/index.htm>



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END of SESSION

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